FireFusion — PRD v2

Challenge-aligned revision with the Ground Node land project Merges your PRD v1 (168 sections), the 2026 challenge statement, and the build guide · Team of 6 · Event 13–14 Nov 2026 · v2.0, 7 Nov 2026

0. What changed from v1, and why (read first)

Your v1 is scientifically the strongest part of the project. Its integrity rules are kept almost word for word (Section 3). But compared with the official challenge statement, v1 has gaps that could cost you at judging. v2 fixes them.

Change 1 — Add a harmonized hotspot time series (the challenge's Part 1). The statement says the first phase is harmonization so that "the sensor of origin no longer matters", producing a consistent series of hotspot activity from the early 2000s to today. v1 deliberately never makes MODIS and VIIRS numerically comparable, and builds only event-level clustering. That leaves the headline deliverable undelivered. v2 keeps your event layer and adds a third layer, the **Harmonized Activity Index (HAI)**: counts of fire-cell-days on a common grid, with the VIIRS-era scale *estimated from real overlap data* (never an invented coefficient), shown with uncertainty and labeled FireFusion-derived. This respects your "no invented NASA coefficients" rule, because a fitted, documented, validated factor is a result, not an invention.

Change 2 — MODIS and long history move from "future work" to MUST. Your supplied package is VIIRS only: NOAA-21 NRT from 2026-01-01 to 2026-10-02, plus small NOAA-20 files from about 2025-10. That cannot produce a calendar "from the early 2000s". v2 makes acquiring MODIS Collection 6.1 standard data (Terra from 2000, Aqua from 2002) and VIIRS S-NPP standard data (from 2012) for your area a first-day task (Section 4.2). Without it, you can only present a 9-month VIIRS demo, and the challenge is about decades.

Change 3 — Scale the event layer to an area. The NOAA-21 file has about 14 million records. Clustering the whole planet in a week is unrealistic and unnecessary. v2 uses the global file for coarse density and statistics, and runs association and clustering only inside the chosen area of interest (default: Bangladesh plus a buffer) and chosen time windows.

Change 4 — Calendar specifics from the challenge. A year-by-month heatmap with an accessible color ramp, the current month compared with the **historical median**, anomalies against multi-year averages, number of hotspots detected, and a downloadable CSV of hotspots for a search. v1 had day/week/month aggregation but not these explicit items.

Change 5 — Restore items you asked for earlier that v1 dropped. A Caution page, a Data and Resources page, a NASA Credits page, prototype watch notifications with an HQ console, and a security section. Blockchain is replaced by a signed hash manifest (a blockchain does not stop anyone hacking a web app; a manifest gives tamper-evidence cheaply).

Change 6 — Add the Ground Node land project as a defined, honest module (Section 9), including corrections to the current blueprint.

Change 7 — Remove the simulated-data fallback (v1 §145) from the submission build. It contradicts "never fabricate". Test fixtures may exist in the test folder only, never in the shipped app.

Change 8 — Branding. Remove the "NASA EARTH OBSERVATION" hero label and any NASA logo. Use "Built for the NASA Space Apps Challenge 2026 · Data from NASA FIRMS" plus a non-endorsement line.

Change 9 — Data inspection finding to check on day one. The NOAA-20 files hold only 2,217 records against 13.99 million for NOAA-21. Confirm whether the NOAA-20 files are a small area or a short period. Do not compare NOAA-20 and NOAA-21 unless both cover the same area and period.

Change 10 — Deployment and timeline are rewritten for the real 13–14 Nov deadline (Section 12).

1. Product statement and positioning

Name: FireFusion. **Title:** Harmonized MODIS & VIIRS Fire Event Intelligence. **Tagline:** One Fire. Multiple Satellites. One Harmonized View.

What it is. A transparent multi-sensor active-fire analytics web app that (A) preserves NASA FIRMS observations with full provenance, (B) groups related observations into FireFusion Events, (C) builds a harmonized, sensor-independent activity series, and (D) presents a Burning Activity Calendar with historical-median comparison for any area and date range.

What it is not. Not a replacement for NASA FIRMS, not an official NASA product, not an operational emergency system, not a validated wildfire forecast.

Presentation statement. FireFusion bridges fragmented satellite fire observations by preserving native MODIS and VIIRS information, evaluating their spatial and temporal relationships, estimating a documented common activity scale from the sensors' overlap,

and presenting unified events and an interpretable Burning Activity Calendar. Every derived result shows how it was produced.

2. Challenge alignment matrix

- **Harmonize MODIS and VIIRS hotspots so sensor of origin no longer matters** → Harmonized Activity Index (Section 5.3) plus event layer. MUST.
- **Web application taking harmonized data as input** → React front end over the API. MUST.
- **Select a region and date range** → draw polygon or box, enter latitude/longitude, predefined regions, current map extent. MUST.
- **Calendar of hotspots from the early 2000s to present** → year-by-month heatmap on harmonized history. MUST.
- **Compare current month with historical median** → default baseline is the median of the same calendar month. MUST.
- **Which months and years had the most activity** → ranked months/years panel. MUST.
- **Anomalies against multi-year averages** → median and mean baselines, percentile and robust z-score. MUST.
- **Color ramps for similarity and difference** → colorblind-safe sequential and diverging ramps. MUST.
- **Downloadable CSV of hotspots for a search** → export with filters, provenance and citation header. MUST.
- **Download analytics and visualizations** → CSV of calendar values, PNG of charts. SHOULD.
- **Audience: early warning, emergency responders, scientists, land managers** → personas, Caution page, watch notifications. SHOULD.
- **Interactive map, latitude/longitude entry, accessible colors** → covered above. MUST.

3. Scientific integrity rules (carried over from v1, non-negotiable)

1. **Ownership labels.** Every metric is tagged NASA FIRMS (source field or product metadata) or FireFusion-derived. Association score, events, activity levels, HAI, calendar and anomalies are always labeled FireFusion-derived.

2. **No fabricated data.** No invented MODIS or VIIRS records, FRP, confidence, view angle, perimeters, live feeds or citations. If MODIS is not loaded, the UI says MODIS NOT LOADED.
3. **Native fields preserved.** MODIS confidence is 0–100 with NASA classes low (below 30), nominal (30 to below 80), high (80 to 100). VIIRS confidence is categorical (l, n, h) and is never converted to a fake percentage. A separate derived quality rank (1, 2, 3) may exist and is never called NASA confidence.
4. **FRP stays raw.** No invented FRP coefficients and no claim that sensor FRP values are equal. Negative FRP is retained, flagged INVALID_FRP and excluded from FRP calculations only.
5. **Nothing silently deleted.** Data-quality outcomes are RETAINED, RETAINED WITH FLAG, EXCLUDED FROM CALCULATION or DEDUPLICATED.
6. **NRT versus Standard.** Provenance kept on every record. Historical baselines prefer Standard; NRT fills the recent gap and is visibly distinguished.
7. **No accuracy percentages** without an independent reference dataset, a defined protocol and proper metrics. The association score and HAI fit statistics are not “accuracy”.
8. **Anomaly is not prediction.** Say Prototype Anomaly Detection. It describes unusualness against the available baseline and does not forecast fires.
9. **Limits always stated:** no detection is not no fire (clouds, overpass timing, small fires); event duration is inferred from overpasses; event footprints are not perimeters; summed FRP is not total energy; detections include non-vegetation thermal sources unless a Type field says otherwise.
10. **Approved wording:** “NASA FIRMS active-fire observations”, “FireFusion Prototype Association Score”, “Prototype Anomaly Detection”, “satellite-detected burning activity”, “FireFusion Event”.
11. **Source hierarchy:** official NASA FIRMS docs, then NASA Earthdata guides, then peer-reviewed research, then FireFusion methodology.

The one rule above all: never make the UI look more certain than the data is. Unknown shows as unknown, absent fields show Not available.

4. Data requirements

4.1 Current package (truth as supplied)

- Four download bundles: J1V-C2 (NOAA-20) ×2 and J2V-C2 (NOAA-21) ×2. NOAA-21 NRT CSV: 13,990,086 records, 2026-01-01 to 2026-10-02, about 1.17 GB. NOAA-20: 35 + 793 + 1,389 records across NRT and archive shapefiles. Raw total 13,992,333; 30 exact cross-file duplicates, so 13,992,303 after deduplication. The app must recompute these numbers at runtime, not hard-code them.
- Known issues: 13 NOAA-21 records with negative FRP (flag, do not delete).
- VIIRS only. No MODIS. The current package supports a recent VIIRS demonstration, not a decades-long calendar.

4.2 Data to acquire (day-one task list, by priority)

1. **MODIS Collection 6.1 Standard (MODIS_SP), Terra and Aqua**, for your primary area, from 2000/2002 to the latest available. Preferred long history.
2. **VIIRS S-NPP Standard (VIIRS_SNPP_SP)** from 2012 for the same area. This is what makes the 2012-onward overlap with MODIS possible.
3. **VIIRS NOAA-20 Standard** (from 2018) and NOAA-21 Standard (from 2024) for the same area, so the VIIRS family is continuous.
4. **MODIS NRT and VIIRS NRT** for the recent gap that Standard does not yet cover.
5. **MCD64A1 burned area** (optional validation layer).
6. **Administrative boundaries** (geoBoundaries or GADM) and a protected-areas layer for the area.

Primary area of interest for the history: Bangladesh plus roughly a 100 km buffer, which also catches cross-border agricultural burning. The archive request tool limits request size, so download in yearly chunks per sensor. Keep the global NOAA-21 file for coarse global density only.

Check dates at ingestion, do not hard-code them. NASA lists Terra MODIS from Nov 2000, Aqua from Jul 2002, VIIRS S-NPP from 2012, NOAA-20 from 2018, NOAA-21 from 2024. The app reads actual min and max dates from the loaded data and shows its local coverage separately from NASA product availability.

4.3 Normalized schema (sensor-neutral)

observation_id, sensor, satellite (Terra, Aqua, SNPP, N20, N21), instrument, latitude, longitude, timestamp (UTC), acquisition_date, acquisition_time, brightness_primary, brightness_secondary, frp_mw, confidence_native, confidence_percent (MODIS only), confidence_class, scan_km, track_km, day_night, product, version, processing_state (NRT, STANDARD, RT, URT), hotspot_type (only where the source has it), source_file, source_format, quality_flag, event_id, grid_cell_id, persistent_source_flag.

Field mapping. MODIS: brightness to brightness_primary, bright_t31 retained, confidence copied to percent and classed by NASA thresholds. VIIRS: bright_ti4 to primary, bright_ti5 to secondary, confidence l/n/h to class, percent null.

4.4 Validation and timestamps

Coordinates within range, schema checked at startup, acq_time zero-padded to HHMM and combined with acq_date as UTC with seconds set to 00 (never invented), Bangladesh local date derived as UTC+6 for calendar bucketing. Cross-file exact duplicates detected and logged. Standard replaces NRT for the same source observation, preserving provenance history.

4.5 Storage

Raw files never committed to Git. Pipeline writes Parquet queried through DuckDB. Frontend never receives millions of records.

5. Processing design: three layers

5.1 Layer A — Observations (native, preserved)

Validated, normalized, deduplicated records with provenance. This is the evidence layer and feeds the map, popups and CSV.

5.2 Layer B — FireFusion Events (your v1 method, scoped)

- **Candidates:** time-sorted, spatially indexed (H3 or BallTree). Never an all-pairs comparison. Only pairs within the configured time and distance windows are considered.

- **Parameters are configuration:** SPATIAL_THRESHOLD_KM (starting point informed by MODIS about 1 km and VIIRS about 375 m pixels plus positional tolerance), TEMPORAL_THRESHOLD_MIN (informed by overpass spacing, for example NOAA-21 about 50 minutes ahead of NOAA-20), minimum cluster size, weights. None is presented as scientifically proven; each is tested in a sensitivity analysis reported in the README.
- **FireFusion Prototype Association Score (0–100):** spatial 40, temporal 30, quality 20, optional FRP 10 (weights configurable and shown). The score is displayed with its components and the exact label every time.
- **Clustering:** ST-DBSCAN-style spatial-temporal density clustering, documented, with dynamic IDs FF-000001 and so on. Statuses ACTIVE, RECENT, HISTORICAL are application labels.
- **Event record:** start, end, inferred duration, observation counts by sensor and satellite, peak, mean and range of valid FRP, confidence composition, association statistics, extent, first, peak and last detection, activity level, ground-evidence status (Section 9). Only fields supported by the real composition are shown.
- **Scope:** events are generated for the area of interest and chosen windows (default: last 90 days plus any user-selected range up to a cap), not for the whole planet.

5.3 Layer C — Harmonized Activity Index (new in v2)

Why. Raw counts are not comparable: one MODIS pixel can correspond to many VIIRS points and VIIRS sees smaller, cooler fires. The challenge wants one consistent series.

Unit: Fire-Cell-Day (FCD). One grid cell (about 1 km; choose and document either 0.01° cells or an equal-area 1 km grid) with at least one qualifying detection on one local calendar date, from any satellite of a sensor family. This collapses VIIRS clusters and Terra/Aqua or multi-VIIRS repeat passes into a comparable unit.

Steps

1. **Quality filter (configurable, defaults documented):** exclude low confidence for both families; apply the Type filter where the field exists; keep flagged records visible.
2. **Persistent source flagging:** cells detecting on at least N distinct days in each of at least Y years (parameters configurable, sensitivity tested) are flagged as persistent heat sources (brick kilns and industry in Bangladesh are the likely examples). They are kept and flagged, excluded from default HAI, and toggleable. Never deleted.
3. **Aggregate:** FCD per day, week and month, per area, per family (MODIS family: Terra+Aqua; VIIRS family: S-NPP+NOAA-20+NOAA-21).

4. **Check VIIRS family continuity:** compare S-NPP, NOAA-20 and NOAA-21 FCD where they overlap, so the VIIRS series is internally consistent before it is mapped to MODIS.
5. **Overlap calibration:** in the period where MODIS and VIIRS both exist (2012 onward), fit a mapping from VIIRS FCD to MODIS-equivalent FCD per region and season class (log-linear regression or quantile mapping). Use the MODIS-equivalent scale as the reference so the 2000s record is untouched and Terra/Aqua retirement does not break the series.
6. **Uncertainty:** carry prediction intervals into every VIIRS-era monthly value and show them.
7. **Validation protocol (all results published):** fit on one block of years and test on held-out years; report bias, MAPE and Spearman rank correlation for naive merge versus harmonized; show the handover step shrinking in the before/after chart; report cross-sensor agreement (share of MODIS FCD with a VIIRS FCD in the same or neighboring cell on the same day, and the reverse); sensitivity to filter and persistence parameters. Report whatever you get, including weak results.
8. **Known biases documented:** Terra and Aqua overpass-time drift in recent years (check current NASA notices before wording this), sensor differences by land cover, and NRT versus Standard differences. Run Terra and Aqua separately as a sensitivity check.
9. **Labeling.** HAI is FireFusion-derived. It is not a NASA product and the calibration is a fitted estimate, shown with its fit statistics.

This reconciles v1's refusal to force equality (raw observations and raw FRP stay native) with the challenge's need for a consistent series (the HAI is a separate, documented, uncertainty-bearing layer).

5.4 Calendar and anomaly engine

- **Granularity:** day, week, month, plus a year-by-month heatmap. Default is chosen from the date range; the user can change it.
- **Metrics per period:** HAI (FCD), observation count (labeled not comparable across sensors), event count, total, mean and peak valid FRP, average confidence class, sensor composition, processing state.
- **Baseline:** default is the **median of the same calendar month across baseline years**, excluding the year being evaluated. Mean and standard deviation and percentile rank are secondary options. Minimum 10 baseline years, otherwise the UI shows Insufficient baseline.

- **Outputs:** difference from median in percent, percentile rank, robust z-score (median absolute deviation).
- **Activity levels (distribution-based, never fixed FRP thresholds):** LOW below the 50th percentile, MODERATE 50th to 75th, HIGH 75th to 90th, VERY HIGH above the 90th. Cutoffs configurable and displayed.
- **Interaction:** selecting a period updates the map, statistics, event list, FRP chart, sensor composition, confidence summary and anomaly status. Area selection recomputes everything for that area, never relabels global numbers.

6. Web application

Pages. 1 Overview. 2 Fire Map. 3 Burning Activity Calendar (the primary page). 4 Area Analysis. 5 Fire Events with detail and timeline. 6 Harmonization Lab (before and after, sensor of origin toggle). 7 Sensor Comparison. 8 Prototype Anomaly. 9 Watch and HQ Console. 10 Ground Network. 11 Caution. 12 Methodology. 13 Data and Resources. 14 NASA Credits and About.

Overview and data status. Hero: FireFusion, the tagline, and a data status strip generated from the real loaded sources, for example NASA FIRMS DATA — LOADED, VIIRS NOAA-20 — LOADED, VIIRS NOAA-21 — LOADED, MODIS — NOT LOADED. The status changes only when validated files are actually loaded. Buttons: Explore Fire Map, How It Works, Launch Demo. No NASA logo and no “NASA Earth Observation” label.

Area of interest. Draw a polygon or rectangle, enter latitude and longitude with a radius, choose a predefined region (country, division, district, protected area), or use the current map extent. Limit vertices and area for performance. The area view always shows which sensors, satellites, processing states and dates actually contribute, so users never assume data that is not there.

Filters (all must work). Date range, harmonized or native sensor view, sensor family, satellite, confidence class, day or night, FRP range, processing state (NRT or Standard), hotspot type (only where the field exists), include or exclude persistent heat sources, and ground-evidence status. Apply and Reset buttons. Filter state lives in the URL so views can be shared.

Map. Leaflet with OpenStreetMap, always backed by API data. Global view shows server-side density grids. Regional view shows server-filtered clustered markers (canvas rendering). A selected event shows its individual observations. Legend: MODIS orange-red, VIIRS purple, FireFusion Event amber, selected cyan outline, ground nodes green

square. Every layer also has an icon or label, so color is never the only cue. Observation popups show only real fields: ID, sensor, satellite, UTC time, coordinates, brightness, FRP, confidence (native), scan and track, day or night, version, processing state, source file, quality flag, event ID.

Burning Activity Calendar.

- Year-by-month heatmap of the HAI over the full available record, with a perceptually uniform, colorblind-safe sequential ramp (viridis family) and a legend.
- Toggle between count and anomaly view; the anomaly view uses a diverging colorblind-safe ramp centered on the median (avoid red versus green).
- Day, week and month views for shorter ranges.
- Hover and focus tooltips: value, rank, uncertainty (for VIIRS-era values), sensors contributing, processing state.
- Ranked panel: months and years with most activity.
- Current month versus historical median card with percent difference, percentile and activity level.
- Table alternative for screen readers, and CSV and PNG export.

Harmonization Lab. Raw MODIS, raw VIIRS, naive merge and harmonized series on one chart, with the handover marked, uncertainty band, and the validation numbers beside it. This is your proof slide. Also the Before and After event view: individual observations on the left (different coordinates, times, FRP, quality), the FireFusion Event on the right (count, composition, duration, extent, peak FRP, association score with its components). Wording: associated observations, grouped under the configured prototype method; never implied to be one physical fire.

Sensor Comparison. Shows only when both families are loaded. Counts, mean and peak FRP, confidence distribution, day and night split, contribution to events, temporal distribution, with the note that the sensors are different instruments. Before MODIS is loaded it shows MODIS COMPARISON UNAVAILABLE and offers VIIRS NOAA-20 versus NOAA-21 only if both cover the same area and period.

Style. Dark mission-control look using your v1 palette (background #07111F, cards #11263A, cyan #00D9FF, MODIS #FF4D3D, VIIRS #A855F7, harmonized #FFB000), Inter font, no cartoon flames, no fake animations, no decorative charts. Contrast checked against WCAG 2.1 AA.

Accessibility and language. Keyboard operation, visible focus, alt text and table equivalents for charts, reduced-motion support, severity shown by color plus icon plus text. Bangla and English toggle (SHOULD).

States. Loading, No Data, Invalid Data, Validation Failed, NASA Data Unavailable, MODIS Not Loaded, API Error, Harmonization Failed, Empty AOI. Never a blank screen.

7. API, architecture and performance

Pipeline. NASA files → ingestion → validation → normalization → deduplication → Parquet → DuckDB → FastAPI → React. Heavy work (grid aggregation, HAI, calibration, events) runs offline or in the API, never in the browser.

Endpoints. Health, provenance, validation, observations and counts, events and event by ID, statistics, calendar, anomalies, areas, harmonize (POST), filter (POST) as in v1, plus: `/api/hai` and `/api/calendar/heatmap`, `/api/export/hotspots.csv`, `/api/export/calendar.csv`, `/api/calibration` (fit and validation statistics), `/api/watch`, `/api/hq/alerts` and acknowledge, `/api/ground/nodes`, `/api/ground/ingest` (POST, signed), `/api/events/{id}/ground`.

CSV export. Native fields, quality flag, processing state, source file, event ID and FireFusion-derived columns clearly labeled, with header comments listing the filters, data versions, and the citation text.

Deployment. Static front end (Vercel or Netlify). API on a small container host (Render, Railway or Fly) with the FIRMS_MAP_KEY and database keys in environment variables only. A precomputed static fallback of the calendar and aggregates ships with the front end, so the demo still works if the API is down. The main demo runs from local validated data and needs no live internet; the FIRMS API is an optional refresh labeled FIRMS Current Data Refresh (never “live feed” unless it truly is).

Performance targets. First view under 3 seconds on 4G, AOI calendar under 2 seconds from precomputed aggregates, no more than a few thousand rendered markers.

Reproducibility. Every harmonization run stores dataset version, source files, configuration, algorithm version, cluster parameters and a timestamp. A methodology debug panel shows the active thresholds, weights, clustering method, minimum cluster size, data window and processing state.

8. Prototype Activity Watch, HQ Console and Caution page

v1 says FireFusion is not an operational emergency system. So notifications are **watch notifications for a research prototype**, never official warnings.

Rules. R1: the area's current-month HAI exceeds the 90th percentile of the same-month history. R2: a new FireFusion Event in a watched area in the last 24 hours with at least N observations (needs NRT data). R3: an event with both sensor families. R4: a ground node alert (Section 9).

Levels. Info, Watch, Elevated, High, each with color, icon and text, mapped to the percentile-based activity levels. Every message states the data source, detection time, satellite revisit limits and the possibility of false detections, and says: prototype watch notification, not an official warning.

Delivery. In-app banner (MUST for this feature), then Telegram bot and email (SHOULD). One alert per cluster per rule per 12 hours, escalation only when the level rises, quiet hours configurable.

HQ Console. Login-protected. Alert queue, map, evidence panel (observations, sensors, association score, ground status), workflow New → Acknowledged → Assigned → Resolved or Marked false, notes, and an append-only audit log. Roles: Viewer, Operator, Admin. A clearly labeled REPLAY mode re-runs a real historical event through the same flow, so the full chain can be demonstrated honestly when no live event occurs.

Caution page. What detections mean and do not mean (cloud, smoke, overpass timing, small fires, non-vegetation heat sources), the current watch level for the selected area, plain safety guidance (smoke exposure, what to do if you see a fire), emergency contacts for Bangladesh (the national emergency number is 999; verify the Fire Service and Civil Defence number from an official source before publishing), and a banner that the prototype does not replace official warnings.

9. Ground Node land project (Sylvan Sphere)

9.1 Role in the system

The Ground Node is an independent, in-situ evidence layer. It does two jobs: (1) attach ground evidence to FireFusion Events and watch alerts, and (2) start the reference data that v1 §140 says future validation needs. It is **not** needed for the calendar, the HAI, or any challenge requirement, so it is built in tiers and never blocks the software.

9.2 Tiers

- **Tier 0 (MUST for submission):** blueprint, data contract, matching workflow, honest limits, and the Ground Network page with an empty state.
- **Tier 1 (SHOULD, one owner, starts today):** one working bench prototype that streams real sensor readings to the API.
- **Tier 2 (only with written permission, can be after the event):** a supervised field pilot.
- **Tier 3 (research extensions):** microbial fuel cell experiments, edge AI, LiDAR, acoustic sensing, satellite uplink.

9.3 Corrections to the current blueprint

- **Power.** A microbial fuel cell yields microwatts to low milliwatts, so it cannot run the electronics. Treat it as a research experiment and log its measured output; never present it as the power source. Power must come from solar plus battery.
- **Edge AI.** A Jetson-class module draws several watts, which defeats a solar sensor node. Use rule-based detection now, and TinyML on a low-power microcontroller later. Say “AI-powered” only after a trained, tested model exists.
- **Encryption.** The flowchart says AES-256. Use the transport’s own encryption (TLS, or LoRaWAN’s built-in AES-128) plus an HMAC signature on every payload, and state exactly that.
- **Fireproof.** Do not claim fireproof or shockproof. Claim only what you test and record how.
- **Scope.** Fauna acoustics, LiDAR tree health and soil-substrate power are separate projects. Remove them from the MVP drawing.
- **Drawings.** Redraw the blueprint yourself with correct labels. The AI-generated images contain garbled text and cannot be your technical evidence.

9.4 Tier 1 bill of materials (buildable in days)

- Microcontroller: ESP32 or ESP32-S3.
- Temperature, humidity and pressure: BME280.
- Particulate matter (smoke indicator): PMS5003 or SPS30. The PMS5003 needs about 30 seconds of fan warm-up per reading, which matters for power.
- Thermal array: MLX90640 (32 by 24 pixels). Its useful range is short, which limits what one node can see.

- Gas: treat low-cost MQ-type sensors as rough indicators only; they drift and cross-react.
- Time and storage: DS3231 real-time clock, microSD for the offline buffer. Optional GPS; otherwise record a surveyed fixed position and the method used.
- Link: cellular module where coverage exists, or Wi-Fi for the bench test. LoRa needs a gateway and you must check the legal band and power rules in Bangladesh before using it. Iridium is Tier 3 (cost).
- Power: Li-ion or LiFePO4 pack with protection, a solar charge controller, a 2 to 6 W panel, a fuse.
- Housing: weatherproof enclosure with a vented, mesh-protected chamber for the particulate sensor.

9.5 Power budget (estimate; measure it with a power meter)

Using typical datasheet currents, an awake cycle of about 40 seconds (microcontroller about 80 mA, thermal array about 23 mA, particulate sensor about 100 mA) averages roughly 27 mA when sampling every 5 minutes, about 90 mW or 2.2 Wh per day. Sampling every 15 minutes averages roughly 9 mA, about 30 mW or 0.7 Wh per day. Two 18650 cells (about 20 Wh) then run about 9 days or 29 days with no sun. A 2 W panel gives about 5 Wh per day in good sun, much less in monsoon cloud or under canopy, so place nodes in clearings or at forest edges and size the panel to the worst season. Replace these numbers with measured ones before publishing them.

9.6 Firmware behavior (simplified from your flowchart)

- Modes by battery: above 50 percent full operation, 20 to 50 percent eco (thermal array off or slower, critical sensors only), below 20 percent deep sleep with status heartbeat.
- Scheduled sampling; sensor range checks; faulty sensor isolated and flagged.
- **Local rule-based alert:** thermal maximum above a calibrated rise over local baseline for k consecutive samples, or a rapid particulate rise, or a temperature rise-rate. Baselines come from at least 48 hours of recorded normal data; thresholds and their derivation are documented.
- Hardware watchdog and hourly self-check; offline FIFO buffer with retry; signed packets with sequence numbers.

9.7 Data contract

Packet fields: node_id, firmware_version, sequence_number, timestamp_utc, latitude, longitude, position_method, temperature, humidity, pressure, pm2_5, pm10, thermal_max, thermal_mean, battery_percent, signal_strength, sensor_fault_flags, alert_flag, source_type (FIELD, BENCH_TEST or SIMULATED_TEST), and hmac. The API rejects unsigned packets, bad signatures, replays (sequence and time window) and malformed values. **Only FIELD packets may be used in any validation statistic.** BENCH_TEST and SIMULATED_TEST packets are displayed with their label and excluded from evidence.

9.8 Matching ground evidence to events

For each FireFusion Event, look for nodes within GROUND_RADIUS_KM (start at 1 km, justified by pixel size plus position tolerance) and alerts within GROUND_WINDOW (start at plus or minus 3 hours around the satellite observations). Both are configurable and tested at other values. Resulting status:

- GROUND_CONFIRMED: a node alert in range and window.
- GROUND_NO_SIGNAL: a node was online in range but raised no alert. This is not a false alarm: small fires, wind and node placement can explain it.
- NO_NODE_IN_RANGE and NODE_OFFLINE: excluded from any rate.
- GROUND_ONLY: a node alert with no satellite event, listed separately.

Report counts, always with sample size. Below roughly 30 matched events present case studies, not rates. A rate or any accuracy-style statement needs an independent reference record (for example forest staff field logs or photographs), a stated protocol and confidence intervals, in line with v1 §139.

9.9 Interface

A Ground Network page with a node map (status, battery, last report, source type), a Ground evidence panel on the event detail page, a ground-evidence filter on the map, and an empty state that says No ground nodes deployed. Ground data never feeds the HAI or the calendar.

9.10 Safety, legal and ethics

- Get written permission from the landowner or forest department before placing anything. Protected areas such as the Sundarbans need official approval.

- Never ignite a fire to test the node. Use a safe, supervised heat source (such as a heat gun at a safe distance) for bench tests, and never leave batteries charging unattended.
- Fuse and protect the battery pack; prefer LiFePO4 for safety.
- Limit soil and wildlife disturbance, mark and remove nodes after the test, and coarsen public node coordinates in sensitive areas.

9.11 Tier 1 done means

The node runs 24 hours continuously, posts signed real readings, appears on the Ground Network page, and a supervised bench heat source triggers a BENCH_TEST alert that flows to the HQ Console, clearly labeled as a bench test.

10. Security requirements

No system is hack-proof; the aim is defense in depth with controls you can list and show.

- **Secrets:** FIRMS_MAP_KEY, database and bot tokens only in server environment variables. Check the repo history for leaks and enable secret scanning.
- **Transport and headers:** HTTPS, HSTS, strict Content Security Policy, frame protection, restrictive referrer policy.
- **API:** rate limits, CORS allowlist, input validation (polygon vertices and area, date ranges), no user-supplied URLs fetched by the server.
- **HQ access:** login with an email allowlist, roles, short sessions, CSRF protection, append-only audit log.
- **Ground packets:** per-node secret, HMAC signature, sequence numbers, time window, schema validation, no over-the-air firmware update in the MVP.
- **Integrity (the useful part of the blockchain idea):** a signed SHA-256 manifest of every published dataset and a daily hash of the audit log, verified in the browser. Anchoring the manifest on a public chain is Phase 3, not MVP.
- **Dependencies:** committed lockfiles, vulnerability alerts, no unused packages.
- **Privacy:** no personal data on public pages; alert recipients stored only for delivery.
- **Threat model page:** assets, threats (tampering, spoofed alerts or packets, key leakage, denial of service) and the control for each.

11. Credits, resources and disclaimers

NASA Credits page. Use the official FIRMS acknowledgment (check current wording on the FIRMS site): data and imagery from NASA's Fire Information for Resource Management System (FIRMS), part of NASA's Earth Science Data and Information System (ESDIS). Also credit the MODIS and VIIRS science teams, LP DAAC for MCD64A1 if used, OpenStreetMap contributors (required attribution on the map), Leaflet and other libraries with licenses, geoBoundaries or GADM, and the Space Apps challenge. State: built for the NASA Space Apps Challenge 2026 and not endorsed by NASA. Add the AI-use disclosure the rules require.

Data and Resources page. For every dataset: name, provider, version, date range loaded, license, link, what it was used for, processing steps, file manifest with hashes, and a changelog. Links to verify before publishing: https://firms.modaps.eosdis.nasa.gov/ , https://firms.modaps.eosdis.nasa.gov/download/ , https://firms.modaps.eosdis.nasa.gov/api/ , https://lpdaac.usgs.gov/products/mcd64a1v061/ , https://www.geoboundaries.org/ , https://gadm.org/ , https://colorbrewer2.org/ , https://www.spaceappschallenge.org/ . Include NASA Earthdata user guides and the peer-reviewed MODIS/VIIRS comparison and fire-clustering papers named in v1's reference list, only after you confirm each citation is real.

UI disclaimers (from v1, keep verbatim in spirit). Scientific disclaimer near the methodology; anomaly text; association score text; MODIS and VIIRS text; current-data text. Add: the Harmonized Activity Index is FireFusion-derived and its calibration is an estimate shown with its fit statistics.

12. Team and timeline (7–14 Nov)

Roles (6).

1. **Data engineer:** ingestion, validation, normalization, Parquet and DuckDB, the MODIS and S-NPP history downloads.
2. **Science and algorithm lead:** FCD, persistent sources, calibration, validation numbers, association score, clustering, sensitivity analysis.
3. **Backend, security and alerts (integration owner):** API, exports, watch rules, HQ Console, security controls, deployment.
4. **Front end, map and area tools:** Leaflet map, area drawing, filters, URL state, event pages.

5. **Front end, visualization and accessibility:** calendar, charts, color ramps, WCAG checks, Bangla, Credits, Resources and Caution pages.
6. **Ground Node and pitch lead:** hardware, firmware, packet client, Ground Network content, slides, video.

Schedule

- **Sat 7 Nov:** data inspection report; start MODIS and S-NPP history downloads; freeze this PRD; assign page owners; confirm the hardware parts are in hand.
- **Sun 8:** loader, validator and normalizer to Parquet; node reading sensors on a breadboard.
- **Mon 9:** FCD, persistent flags and monthly aggregates; calendar on real data; node logging and posting to the API.
- **Tue 10:** calibration and validation numbers; area API, CSV export, map; node enclosure and power test.
- **Wed 11:** events on the area window; anomaly view; watch rules and HQ page; ground evidence panel; Credits, Resources, Caution pages.
- **Thu 12:** sensitivity analysis, security checklist, tests, **feature freeze**; 24-hour node run; first full demo rehearsal.
- **Fri 13 – Sat 14:** polish, accessibility pass, video, submission, final rehearsal. Check the national-level rules on pre-built work and disclose what existed before.

If you fall behind, cut in this order: Bangla, Telegram and email, PNG export, Tier 1 node (keep Tier 0), extra event features. Never cut the HAI, the calendar, the median comparison, the CSV or the credits.

13. Testing and acceptance

Tests. Unit tests for timestamp handling, confidence classing, deduplication, FCD aggregation, calibration mapping, association score, clustering on small known inputs, anomaly math, and ground matching; API tests; filter and calendar tests; a data-quality report from the pipeline; accessibility checks (automated plus keyboard); security checklist on the deployed site.

Acceptance — data. Real files discovered and loaded; counts recomputed at runtime; MODIS status shown truthfully; UTC timestamps; FRP and confidence validated; cross-file duplicates found; NRT and Standard preserved.

Acceptance — harmonization. HAI exists for the full loaded record; calibration fit and held-out validation published; before and after chart shows the handover effect; events and scores are generated from data and labeled FireFusion-derived; sensitivity analysis reported.

Acceptance — calendar and area. Year-by-month heatmap with accessible ramp; median comparison with percent difference and percentile; ranked months and years; area selection recomputes everything for that area; CSV downloads with filters and citation.

Acceptance — ground and alerts. Tier 0 artifacts complete; if Tier 1 is built, the 24-hour run and labeled bench alert work; the Ground Network page shows an honest empty state if nothing is deployed.

Acceptance — integrity. No fabricated records, FRP, confidence, view angle, accuracy numbers, predictions, live feeds or citations; NASA versus FireFusion ownership visible; credits and non-endorsement present; no secrets in the repo.

14. Risks and mitigations

- **MODIS history not downloaded in time.** Start Saturday; if it fails, present a VIIRS-only recent calendar labeled honestly, with the pipeline proven on whatever data you have.
- **Calibration looks weak.** Publish it with uncertainty; the transparency is itself a strength.
- **Event layer too slow.** Restrict to the area and window, use the spatial index, precompute.
- **API down at judging.** Static precomputed fallback.
- **Hardware late.** Keep Tier 0; show labeled bench data only if real.
- **Overclaiming.** Review every slide against Section 3.
- **Overload.** Page owners, daily standup, feature freeze on 12 Nov.

15. Submission package and demo

Package. Live link, public repo with a full README (v1 §141 list plus the HAI and Ground Node sections), video, one-page method figure with validation numbers, project description, team, data and AI-use disclosure, credits.

3-minute demo. 0:00 the problem in one sentence. 0:20 data status strip, with honest coverage. 0:40 Harmonization Lab, before and after. 1:10 draw an area, show the calendar and the median comparison. 1:40 open an event, its before and after, timeline and ground

evidence panel. 2:10 CSV download and the Caution page. 2:25 watch notification into the HQ Console (replay, labeled). 2:45 Ground Node concept or prototype and roadmap. 2:55 credits.

Judge answers (v1 versions kept). Why harmonize: sensors differ, so FireFusion keeps native data, relates observations, and adds a documented common activity scale. Is the score NASA's: no, it is FireFusion-derived. Do you predict fires: no, the anomaly describes unusualness against a baseline. Why are FRP values different: sensor and retrieval differences, so FRP stays native. New: **Why is the HAI valid:** it is fitted on real overlap data, validated on held-out years, and shown with uncertainty. **What does the Ground Node prove:** only the counts it records, with sample size; it is a pilot, not a validation of satellite accuracy.

16. Developer instructions (for Claude Code or any coder)

1. **Start with data inspection, not a dashboard:** file discovery, archive and schema inspection, record counts, date and spatial coverage, sensor, satellite, product and processing state, and an internal dataset report. Check the NOAA-20 versus NOAA-21 asymmetry.
2. Confirm definitions from official sources: MODIS C6.1, VIIRS Collection 2, confidence, FRP, scan and track, NRT versus Standard.
3. Build loader, validator, normalizer and deduplicator before any UI polish.
4. Build FCD, persistent flags, HAI aggregates, calibration and validation; then association, clustering and event analytics, with tests.
5. Build calendar, anomaly and area analysis.
6. Build map, charts, timeline, Harmonization Lab, methodology and provenance.
7. Build watch rules, HQ Console, Ground Network and ground matching.
8. Run every test and the demo on real data; add real MODIS history through the same pipeline.

Suggested repository: `pipeline/` (ingest, validate, normalize, dedupe, fcd, calibrate, events, anomaly, manifest, tests), `backend/` (API, exports, watch, hq, ground), `frontend/` (pages, map, charts, services), `firmware/` (node code and packet client), `docs/` (this PRD, threat model, validation report, hardware blueprint), `data/` (samples and schema only).